Caitlin Coffey

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Education

Electrical and Computer Engineering, May 2023, GPA 3.96/4.00
Olin College of Engineering, Needham, MA

- Recipient of a 4-year, 50% Olin Merit Scholarship
- **Relevant coursework:** Data Structures and Algorithms, Software Systems, Software Design, Computer Architecture, Discrete Mathematics, Modeling and Simulation of the Physical World (ModSim), User-Oriented Collaborative Design, Introduction to Sensors and Instrumentation, Quantitative Engineering Analysis
- **Clubs:** Public Interest Technology, Olin Conductorless Orchestra, Society of Women Engineers

Work Experience

The MITRE Corporation, Bedford, MA
Data Science Intern (05/2021)-(08/2021)

- Design, analyze, and publish an automated data pipeline in Apache Airflow and PySpark.
- Conduct research on recurrent neural network designs and debug extensive Python code in Tensorflow, Apache Airflow, and PySpark

Qualcomm Inc., Boxborough, MA
Software Engineering Intern (05/2020)-(08/2020)

- Create a software tool to analyze GPU performance in Python using Matplotlib, NumPy, Tkinter, and Pandas.
- Troubleshoot GPU performance using Linux, Vim, and SystemVerilog.

Olin College of Engineering, Needham, MA
Software Design & ModSim Teaching Assistant (09/2020)-(Current)

- Assist in teaching data analysis/data science concepts. Analyze and debug code in Python and MATLAB.

Volunteer Experience

Code for Boston, Cambridge, MA
Software Developer (12/2019)-(04/2020)

- Build and test a software tool in YAML and Docassemble to help people from low-income backgrounds expunge or seal their criminal records.
- Manage an open-source project on Github with six other developers.

Relevant Skills

Python – Object-oriented programming, artificial intelligence (Markov Chain Analysis, Minimax, Djisktra's Shortest Path, deep neural networks). Extensive experience with Pandas, NumPy, Tensorflow, and Multiprocessing libraries

C/C++ - Reverse engineering (using Ghidra), object-oriented programming (creating a LC-3 Virtual Machine)

MATLAB – Data analysis, artificial intelligence (facial recognition using K-nearest neighbor, principal component analysis, and eigenvalue decomposition algorithms), audio processing (using Discrete Fourier transforms)

Linux – Running and creating C-shell/Bash scripts, automating test scripts, managing processes, version-control navigation (Git/Perforce) and system navigation